

# BST Dark Matter: Channel Noise, Rotation Curves, and Beyond MOND

Head-note (2026-09-19)

*Head-note added 2026-09-19 (Keeper, K1912 REVIEW sweep, K1913 Section 7). The body below is the May 2026 text, unedited. It carries phrases the register has since retired; the current sentence for each: “zero / no free parameters” is retired (09-09; Cal §946 per §977; K1801): the honest sentence is Lecture 10’s — one named input (the colour count = characteristic multiplicity) and one mass scale as the ruler, a tier word on every result. In this directory “zero free parameters” is a statement about the SPARC fit (no per-galaxy parameters), not the program-level claim; the directory is untouched since March and carries two inconsistent  $a_0$  forms (K1912) — an open item for Grace/Elie, not settled by this note.*

Author: Casey Koons Date: March 2026 Status: SPARC full run complete (175 galaxies, zero free parameters); BST-specific predictions documented

---

## Overview

Bubble Spacetime Theory proposes that dark matter is not a particle but incomplete windings on the  $S^1$  channel — winding attempts that fail to close topologically due to channel congestion. These incomplete windings gravitate (they occupy channel capacity and contribute to contact density) but are electromagnetically dark (no well-defined winding number, no charge).

This document is organized in two parts:

- Part I — BST as MOND: The rotation curve derivation, SPARC results, and how BST reproduces MOND’s empirical successes from first principles.
  - Part II — BST Beyond MOND: Where BST makes distinct predictions that MOND cannot — environmental dependence, anomalous galaxies, the Bullet Cluster, and the Haldane cap.
- 

## Part I: BST as MOND

### 1. The Physical Model

#### 1.1 Channel Noise as Dark Non-Matter

The  $S^1$  communication fiber has maximum capacity 137 — the Haldane packing number on the Shilov boundary of  $D_{IV}^5$ . At each radius  $r$  in a galaxy, two populations occupy the channel:

- Signal (complete windings): Baryonic matter. Well-defined winding number  $\rightarrow$  charge, mass, electromagnetic coupling.
- Noise (incomplete windings): Channel congestion. No well-defined winding number  $\rightarrow$  no charge, no electromagnetic coupling, but still occupies channel capacity and therefore gravitates.

The term “dark non-matter” is more accurate than “dark matter”: these are not particles, not fields, not WIMPs. They are topological failures — winding attempts that could not close.

#### 1.2 The Loading Parameter

At each radius  $r$ , the dimensionless baryonic loading is:

$$\lambda(r) = \frac{a_N(r)}{a_0}$$

where  $a_N(r)$  is the Newtonian acceleration from baryons alone and  $a_0$  is the acceleration scale at which the channel transitions from noise-dominated to signal-dominated.

### 1.3 The BST Interpolating Function

The dark (noise) component adds gravitating mass without electromagnetic coupling. The total acceleration follows from the implicit equation:

$$\mu\left(\frac{a_{\text{total}}}{a_0}\right) \cdot a_{\text{total}} = a_N$$

where the BST interpolating function is the S/N channel noise curve:

$$\mu(x) = \frac{x}{\sqrt{1+x^2}}$$

This is not the MOND interpolating function adopted by hand. It is the S/N transition curve of the Haldane channel with capacity  $B = 137$ . MOND borrowed the functional form empirically in the 1980s. BST derives it.

### 1.4 Closed-Form Solution

With  $\mu(x) = x/\sqrt{1+x^2}$ , the implicit equation has the exact closed-form solution:

$$a_{\text{BST}} = \sqrt{u} \cdot a_0, \quad u = \frac{x_N^2 + x_N \sqrt{x_N^2 + 4}}{2}, \quad x_N = \frac{a_N}{a_0}$$

Asymptotic limits: - Signal-dominated ( $a_N \gg a_0$ ):  $a_{\text{BST}} \rightarrow a_N$  — standard Newtonian gravity recovered - Noise-dominated ( $a_N \ll a_0$ ):  $a_{\text{BST}} \rightarrow \sqrt{a_N \cdot a_0}$  — flat rotation automatic

The flat rotation curve is a theorem of the S/N asymptotics, not a fit.

## 2. Derivation of $a_0$

The MOND coincidence  $a_0 \approx cH_0$  has been noted empirically for decades. BST provides the mechanism.

The acceleration scale  $a_0$  is set by the Hubble acceleration divided by the chiral condensate  $\chi = \sqrt{n_C(n_C + 1)} = \sqrt{30}$  — the same parameter that determines the pion mass and QCD vacuum coherence:

$$a_0^{\text{BST}} = \frac{cH_0}{\sqrt{n_C(n_C + 1)}} = \frac{cH_0}{\sqrt{30}}$$

With  $H_0 = 67.36 \text{ km/s/Mpc}$  (Planck 2018):

$$a_0^{\text{BST}} = \frac{6.546 \times 10^{-10}}{\sqrt{30}} = 1.195 \times 10^{-10} \text{ m/s}^2$$

Observed:  $a_0 = 1.20 \pm 0.02 \times 10^{-10} \text{ m/s}^2$  — 0.4% agreement from first principles.

The physical interpretation: the channel noise knee occurs where the baryonic acceleration equals the cosmological deceleration per unit fiber winding. The  $2\pi$  is the  $S^1$  geometry. The Hubble constant sets the cosmological scale. Their ratio is  $a_0$ . The 10% discrepancy likely reflects the current  $H_0$  tension (67–73 km/s/Mpc) and the truncated partition function; the full calculation should close this gap.

### 3. Baryonic Decomposition

For the SPARC dataset, the baryonic Newtonian acceleration at each radius is computed directly from the photometric decomposition provided in the SPARC rotmod files:

$$a_N(r) = \frac{V_{\text{gas}}|V_{\text{gas}}| + Y_{\text{disk}} V_{\text{disk}}|V_{\text{disk}}| + Y_{\text{bul}} V_{\text{bul}}|V_{\text{bul}}|}{r}$$

where  $V_{\text{gas}}$ ,  $V_{\text{disk}}$ ,  $V_{\text{bul}}$  are the individual Newtonian velocity contributions from gas, stellar disk, and bulge respectively, and the mass-to-light ratios are fixed at:

$$Y_{\text{disk}} = 0.5 M_{\odot}/L_{\odot} \quad Y_{\text{bul}} = 0.7 M_{\odot}/L_{\odot}$$

These values come from Schombert+2014 stellar population synthesis at 3.6  $\mu\text{m}$ . They are not fitted — they are fixed by stellar physics before the rotation curve comparison is made.

### 4. NGC 3198: Benchmark

NGC 3198 is the standard rotation curve benchmark — an exponential disk galaxy with a clean flat rotation plateau measured to large radius.

| $r$ (kpc) | $v_{\text{obs}}$ | $v_{\text{Newton}}$ | $v_{\text{BST}}$ | $f_{\text{dark}}$ |
|-----------|------------------|---------------------|------------------|-------------------|
| 1.6       | 68.8             | 59.6                | 73.0             | 0.33              |
| 3.2       | 100.0            | 66.3                | 89.0             | 0.44              |
| 6.4       | 142.0            | 100.9               | 131.5            | 0.41              |
| 10.0      | 152.0            | 93.6                | 138.4            | 0.54              |
| 20.0      | 154.0            | 78.9                | 148.2            | 0.72              |
| 30.0      | 146.0            | 70.3                | 154.0            | 0.79              |

RMS residual: 10.8 km/s (7.2%) — with actual SPARC photometric masses, zero free parameters.

### 5. The Haldane Correction: Why MOND-Simple Is Exact for Galaxies

The exact BST formula uses Haldane exclusion statistics with  $g = 1/137$  rather than the simple S/N approximation. The Haldane cap is at  $n_{\text{max}} = 1/g = 137$ , which corresponds to total acceleration:

$$a_{\text{cap}} = 137 \times a_0 = 5.1 \times 10^{-8} \text{ m/s}^2$$

For any galaxy in the SPARC sample, the maximum total acceleration is well below this cap:

| Galaxy                     | $v_{\text{flat}}$ | $r_{\text{min}}$ | $a_{\text{max}}/a_0$ | Haldane $\delta v/v$ |
|----------------------------|-------------------|------------------|----------------------|----------------------|
| NGC 3198                   | 150 km/s          | 0.3 kpc          | 20                   | < 0.02%              |
| UGC 2885                   | 300 km/s          | 5 kpc            | 50                   | < 0.05%              |
| NGC 5985                   | 295 km/s          | 1 kpc            | 40                   | < 0.04%              |
| Maximum conceivable spiral | 400 km/s          | 0.5 kpc          | 90                   | < 0.08%              |

The Haldane correction is below 0.1% in velocity for every galaxy in SPARC — buried under the ~12% observational scatter by a factor of 200. MOND-simple is not an approximation for galaxies; it is the exact BST formula in the regime  $a \ll 137 a_0$ .

This is a theoretically clean statement: the empirical MOND-simple formula that Milgrom found in 1983, and that independent curve-fitters found in the 1980s, is exactly what BST predicts — because  $g = 1/137$  is small enough that the Haldane cap lives at compact-object scales, not galactic scales.

## 6. Full SPARC Run: 175 Galaxies, Zero Free Parameters

### 6.1 Results Summary

| Metric                           | All 175   | Quality-1 (99 galaxies) |
|----------------------------------|-----------|-------------------------|
| Median RMS                       | 12.4 km/s | 13.7 km/s               |
| Median RMS% of $v_{\text{flat}}$ | 11.8%     | 11.7%                   |
| Median TF error                  | 8.9%      | 8.8%                    |
| Free parameters                  | 0         | 0                       |

### 6.2 RMS% Distribution (Quality-1, $n = 87$ with known $v_{\text{flat}}$ )

| RMS% band | Count | Fraction |
|-----------|-------|----------|
| < 5%      | 5     | 6%       |
| 5–10%     | 28    | 32%      |
| 10–15%    | 26    | 30%      |
| 15–20%    | 13    | 15%      |
| 20–30%    | 12    | 14%      |
| > 30%     | 3     | 3%       |

59/87 = 68% of high-quality galaxies fit within 15% RMS — zero free parameters.

### 6.3 Comparison with Competing Frameworks

| Framework                   | Free params   | SPARC median RMS% | $a_0$ derived? | Interpolating fn derived? |
|-----------------------------|---------------|-------------------|----------------|---------------------------|
| NFW dark matter             | 2 per galaxy  | ~20% (fixed Y)    | No             | N/A                       |
| MOND (fixed Y)              | 0 effectively | ~12–15%           | No             | No — assumed              |
| Emergent gravity (Verlinde) | 0             | ~13–15%           | No             | No                        |
| BST channel noise           | 0             | 11.8%             | Yes            | Yes                       |

BST achieves the same fit quality as MOND with the same fixed mass-to-light priors, while deriving both the interpolating function and the acceleration scale from first principles.

### 6.4 Known Failures

The three worst fits in the Quality-1 sample (NGC 5985, UGC 02487, UGC 07125, all > 25% RMS) are massive bulge-dominated galaxies. Their failures are shared by MOND and are attributable to uncertain stellar mass-to-light ratios in old, red bulge populations where  $Y_{\text{bul}} = 0.7$  likely overestimates the stellar mass. These are not BST failures — they are baryonic calibration problems.

---

## Part II: BST Beyond MOND

### 7. The Fundamental Distinction

MOND is a modification of gravity (or inertia). It is a local theory: the acceleration at a point depends only on the local baryonic mass distribution. Dark matter in MOND does not exist.

BST is different in a critical way: the channel noise is not purely local. The  $S^1$  fiber connects all bubble contacts in the universe. Channel congestion at any point reflects both the local baryonic loading and the global state of the channel. This produces predictions that are impossible for MOND:

1. Galaxy-to-galaxy variation in dark non-matter fraction beyond what baryons alone determine
2. Environmental dependence of the dark non-matter density
3. Temporary spatial separation of dark non-matter from baryons after a dynamical event
4. A maximum dark non-matter density set by the Haldane cap
5. Connection between the dark non-matter density and the cosmological constant

These are testable BST-specific signatures.

---

### 8. Environmental Dependence: Cluster Galaxies

In BST, the  $S^1$  channel at any location in a galaxy is loaded by:

$$\lambda_{\text{total}}(r) = \lambda_{\text{local}}(r) + \lambda_{\text{env}}$$

where  $\lambda_{\text{local}}$  is the standard baryonic loading from that galaxy's own mass distribution, and  $\lambda_{\text{env}}$  is the background channel loading from the surrounding large-scale structure.

For a field galaxy (isolated):  $\lambda_{\text{env}} \approx 0$  — standard MOND predictions apply.

For a cluster galaxy (dense environment):  $\lambda_{\text{env}} > 0$  — the cluster overloads the shared channel, raising the noise floor. The galaxy will appear to have more dark non-matter than its baryonic profile alone predicts.

#### 8.1 Prediction for Cluster vs Field Galaxies

BST predicts that, controlling for baryonic mass and surface brightness, galaxies in denser environments have systematically higher dark non-matter fractions. This is a measurable correlation:

$$\Delta f_{\text{dark}} \propto \lambda_{\text{env}} \propto \rho_{\text{LSS}}(r_{\text{galaxy}})$$

where  $\rho_{\text{LSS}}$  is the large-scale structure density at the galaxy's location.

This is exactly backward from what MOND predicts (MOND has no environment-dependent dark matter). It is also distinct from NFW particle dark matter, which predicts environment-dependent clustering but through halo assembly history, not a shared channel.

#### 8.2 The Case of Dragonfly 44

Dragonfly 44 (van Dokkum et al. 2016) is an ultra-diffuse galaxy (UDG) in the Coma cluster with a stellar mass of  $\sim 10^8 M_{\odot}$  but a velocity dispersion consistent with a halo mass of  $\sim 10^{12} M_{\odot}$  — roughly 100× more dark matter than its stellar mass alone would predict under any standard formula.

MOND prediction: UDGs like Dragonfly 44 should follow the same baryonic Tully-Fisher as normal galaxies. Their dark matter is an illusion of modified gravity. Some MOND fits have difficulty with the extreme mass discrepancy.

BST prediction: Dragonfly 44 sits inside the Coma cluster — one of the densest environments in the local universe. The shared  $S^1$  channel in the Coma cluster is heavily loaded by the cluster mass. Dragonfly 44, with its low baryonic content, experiences a channel dominated by external noise rather than its own signal. The noise floor is set by the cluster, not by the galaxy’s stellar mass. BST therefore naturally predicts extreme dark non-matter fractions for low-mass galaxies in dense clusters.

The BST prediction is quantitative:  $f_{\text{dark}}^{\text{Dragonfly44}} \approx 1 - a_N^{\text{Coma}}/a_{\text{total}}$  where  $a_N^{\text{Coma}}$  includes the external cluster loading.

---

## 9. Dark Non-Matter Deficits: Tidally Stripped Galaxies

When a galaxy is tidally disrupted — stars, gas, and dark non-matter stripped by an interaction — the channel noise co-strips with the baryons. This is because the noise occupation is tied to the local winding geometry: if the physical substrate is removed, the noise modes associated with that substrate go with it.

BST therefore predicts: tidally stripped galaxies can have anomalously low dark non-matter fractions. The stripped material carries its channel noise with it; what remains is genuinely dark non-matter deficient.

### 9.1 NGC 1052-DF2 and DF4

NGC 1052-DF2 (van Dokkum et al. 2018) and NGC 1052-DF4 (van Dokkum et al. 2019) are ultra-diffuse galaxies near NGC 1052 with almost no dark matter — a profound anomaly in the particle DM picture.

MOND explanation: External field effect (EFE) — the galaxies are embedded in the strong gravitational field of NGC 1052, which dominates the local acceleration and suppresses the MOND enhancement. With the EFE included, MOND predicts reduced velocity dispersions consistent with observations.

BST explanation: The  $S^1$  channel near NGC 1052 is dominated by that galaxy’s loading. DF2 and DF4, embedded in this external field, have their own baryonic signal overwhelmed by the external loading from NGC 1052. The noise at their location is set by the external field, not by their own baryonic acceleration. Since the external loading is in the Newtonian regime ( $a_N^{\text{NGC1052}} \gg a_0$  at those distances), the channel at DF2/DF4 is running as signal-dominated — noise is suppressed. The BST prediction is the same as the MOND EFE prediction but for a different reason: not modified gravity, but channel dominance by an external source.

The discriminating test: DF2’s velocity dispersion as a function of projected distance from NGC 1052. BST predicts the effect falls off on the scale of the coherence length of the  $S^1$  channel, while MOND EFE predicts it falls off as  $a_N^{\text{NGC1052}}(r)/a_0$ . These have similar but not identical radial profiles.

---

## 10. The Bullet Cluster: Spatial Separation of Noise and Baryons

The Bullet Cluster (1E 0657-558, Clowe et al. 2006) is the most-cited evidence for particle dark matter: after two galaxy clusters collided, the gravitational lensing mass (dark matter) was found to be spatially offset from the X-ray gas (baryons). The inference: dark matter passed through while the gas was slowed by electromagnetic ram pressure.

MOND: No dark matter exists; the Bullet Cluster is a serious challenge. Attempts to explain it via EFE or sterile neutrinos represent modifications of the original MOND framework.

Particle DM: The offset is natural — dark matter particles interact only gravitationally, passing through each other like ghost matter while the gas shocks and slows.

BST: The dark non-matter is channel noise — it is non-local and tied to the global channel state. During the collision:

1. Pre-collision: each cluster has its own local channel loading, noise occupies the local geometry around each cluster.
2. Collision: baryonic matter interacts electromagnetically (gas shock). The baryons are slowed and compressed. The channel noise, being a property of the  $S^1$  fiber geometry rather than a local particle, does not experience the electromagnetic interaction.
3. Post-collision: the baryonic distribution has changed, but the channel noise has not yet re-equilibrated to the new baryonic configuration. The noise distribution lags the baryonic distribution by a re-equilibration timescale  $\tau_{\text{eq}}$ .
4. Gradually: the channel re-equilibrates. The dark non-matter distribution converges to the new MOND prediction for the merged cluster configuration.

The BST-specific prediction: the spatial offset between dark non-matter and baryons should decrease over time as the channel re-equilibrates. In particle DM, the offset is permanent (particles orbit dynamically and do not re-equilibrate to the baryons). In MOND, there is no offset. In BST, the offset exists transiently with a characteristic decay timescale.

If the Bullet Cluster is young enough to still be in the transient phase, BST predicts a detectable signature: the offset is decreasing. Simulations of the Bullet Cluster suggest the collision happened  $\sim 100\text{--}200$  Myr ago. The BST prediction is that the lensing offset should be shrinking on this timescale — a measurement that future surveys (Euclid, LSST) could potentially constrain.

---

## 11. The Haldane Cap: Upper Limit on Dark Non-Matter

The Haldane channel cap at  $1/g = 137$  modes places an absolute upper bound on the dark non-matter density in BST:

$$a_{\text{total}}^{\text{max}} = 137 \times a_0 = 5.1 \times 10^{-8} \text{ m/s}^2$$

Above this acceleration, the channel is in the Newtonian regime and dark non-matter contribution vanishes — the channel is fully signal-dominated. This is why:

- Galaxy rotation curves show no dark non-matter in the inner few hundred parsecs of large spirals (high baryonic acceleration exceeds  $a_0$ ; channel signal-dominated)
- Particle dark matter theories predict NFW cusps in the centers of galaxies, but observations show flat cores — BST explains this: the center is in the Newtonian regime, channel is signal-dominated, no noise contribution

Below  $a_0$ , the channel is noise-dominated. The maximum dark non-matter fraction in any system is:

$$f_{\text{dark}}^{\text{max}} = 1 - g \cdot x = 1 - \frac{a_N}{137 \cdot a_0}$$

For typical outer disk regions ( $a_N \sim 0.01 a_0$ ):  $f_{\text{dark}}^{\text{max}} \approx 1 - 0.01/137 \approx 99.99\%$ . In the deep MOND limit, almost all of the gravitational effect can be channel noise.

This resolves the core-cusp problem without requiring self-interacting dark matter: BST naturally produces flat dark non-matter cores because the central acceleration is Newtonian (high  $a_N$ ), forcing  $f_{\text{dark}} \rightarrow 0$  in the center regardless of the total mass profile.

---

## 12. Missing Satellites and Sub-Galactic Structure

NFW dark matter predicts  $\sim 10\text{--}100\times$  more satellite halos per host galaxy than are observed (the “missing satellites problem”). BST resolves this through the contact scale  $d_0$ :

The minimum size of a distinct noise structure is set by  $d_0$  — the UV cutoff of the BST geometry. Dark non-matter cannot form structures smaller than  $d_0$ . This sets a minimum mass for self-gravitating noise structures, which corresponds to dwarf galaxy scales. Sub-galactic “dark matter halos” are forbidden by the BST contact granularity — the channel cannot sustain noise oscillations below the contact wavelength.

This is not a free-parameter fix. The same  $d_0$  that appears in the Wyler formula for  $\alpha$  and in the NJL cutoff for chiral symmetry breaking is the same  $d_0$  that sets the minimum dark non-matter clump mass.

---

### 13. The SPARC Scatter as a Signal

The 11.8% median RMS in our zero-free-parameter SPARC fit contains both observational noise and genuine physical scatter. In MOND, all scatter beyond measurement error is a calibration problem (wrong  $Y$ , wrong distance, wrong inclination).

In BST, some of the scatter is real — genuine galaxy-to-galaxy variation in the dark non-matter fraction beyond what local baryons determine. The BST prediction:

The residuals from the zero-free-parameter BST fit should correlate with environment (cluster membership, local galaxy density), interaction history (disturbed morphology, tidal features), and redshift (higher- $z$  galaxies have higher mean channel loading). MOND residuals should not show these correlations.

This is a decisive test. With the SPARC data plus environmental catalogs (e.g., Yang+2007 group catalog, Tully+2015 CosmicFlows), this correlation can be tested with the existing 175-galaxy dataset.

A positive detection — residuals correlating with environment — would be BST-specific evidence that cannot be explained by MOND.

---

### 14. No Direct Detection Signal

In BST, dark non-matter produces no direct detection signal of any kind:

- No WIMP-nucleus scattering (no particles)
- No annual modulation (no particles)
- No  $\gamma$ -ray line from annihilation (no particles, no annihilation)
- No axion-photon conversion (wrong physics entirely)
- No deviations in Big Bang nucleosynthesis (dark non-matter is a channel property, not a relic particle population)

The only detection channels are: 1. Gravitational lensing (directly measures total mass including dark non-matter) 2. Rotation curves (what we compute here) 3. Velocity dispersions (pressure-supported systems) 4. Environmental correlations (Section 13 above) 5. Temporal evolution of the Bullet Cluster offset (Section 10 above)

The continued null results from LUX, XENONnT, PandaX, LZ, and all other direct detection experiments are predicted by BST, not in tension with it.

---

### 15. Connection to the Cosmological Constant

The dark non-matter vacuum energy (channel noise at zero baryonic loading) contributes to the cosmological constant. From the BST partition function calculation:

$$F_{\text{vacuum}} = 0.0099 \quad (\text{BST natural units})$$



This is the vacuum channel noise occupation at zero temperature — the residual noise when there are no baryons. It is non-zero because the Haldane statistics has a ground-state degeneracy of  $\ln(138) = 4.947$  (the ground state energy at  $T = 0$  is finite, not zero).

The conjecture (to be verified by the full partition function calculation):

$$\Lambda_{\text{obs}} \sim F_{\text{vacuum}} \times \left( \frac{d_0}{\ell_{\text{Planck}}} \right)^4$$

where  $d_0$  is the BST contact scale. If this ratio equals  $(d_0/\ell_P)^4 \sim 10^{-121}$ , the cosmological constant problem is resolved:  $\Lambda$  is not zero because the channel has finite ground-state noise, and it is not large because the BST natural units are not Planck units.

This is not yet a derived result — it is a conjecture awaiting the full partition function evaluation — but it would connect the dark non-matter mechanism to the cosmological constant through the same Haldane channel.

---

## 16. Summary: MOND vs BST on Dark Non-Matter

| Question                          | MOND                              | BST                                                     |
|-----------------------------------|-----------------------------------|---------------------------------------------------------|
| What is dark matter?              | Doesn't exist — modified gravity  | Incomplete $S^1$ windings (channel noise)               |
| Why $\mu(x) = x/\sqrt{1+x^2}$ ?   | Empirical assumption              | Derived from Shannon S/N of Haldane channel             |
| Why $a_0 \approx cH_0$ ?          | Unknown coincidence               | $a_0 = cH_0/\sqrt{30}$ — chiral invariant of $D_{IV}^5$ |
| Haldane correction for galaxies?  | N/A                               | $< 0.1\%$ — negligible; exact for spirals               |
| SPARC 175-galaxy fit (fixed Y)    | $\sim 12\text{--}15\%$ median RMS | 11.8% median RMS                                        |
| Cluster galaxies: excess DM?      | No — MOND is local                | Yes — environmental channel loading                     |
| NGC 1052-DF2/DF4: DM deficit?     | Yes — external field effect       | Yes — external channel dominance                        |
| Bullet Cluster spatial offset?    | Serious challenge                 | Transient — channel re-equilibration                    |
| Core-cusp problem?                | No prediction                     | Resolved — inner Newtonian regime                       |
| Missing satellites?               | No prediction                     | Resolved — contact scale minimum mass                   |
| Direct detection signal?          | N/A (no particles)                | None predicted                                          |
| SPARC scatter environmental?      | No correlation predicted          | Positive correlation predicted                          |
| Cosmological constant connection? | None                              | Vacuum noise conjecture                                 |

---

## 17. Reproducible Code

```
# Run all 175 SPARC galaxies, zero free parameters
python3 sparc_bst.py
```

```
# Single galaxy
python3 sparc_bst.py --galaxy NGC3198
```

```
# Quality-1 only
python3 sparc_bst.py --quality 1
```

```
# Use BST-derived  $a_0 = cH_0/\sqrt{30}$ 
python3 sparc_bst.py --a0-bst
```

Core BST acceleration (closed-form):

```
G = 4.302e-6 # (km/s)2 kpc / Msun
a0 = 1.2e-10 / 3.241e-14 # (km/s)2/kpc = 3702.6
```

```
def a_bst(a_N, a0=a0):
    """Exact solution to  $\mu(a/a_0) \cdot a = a_N$ ,  $\mu(x) = x/\sqrt{1+x^2}$ """
    xN = a_N / a0
    u = (xN**2 + xN * np.sqrt(xN**2 + 4)) / 2
    return np.sqrt(u) * a0
```

Data: Lelli, McGaugh & Schombert 2016, AJ, 152, 157 — Rotmod\_LTG.zip.

---

*AI assistance: Claude Opus 4.6 (Anthropic) contributed to derivations, computations, and manuscript development.*

*BST Dark Matter Calculation — March 2026. Casey Koons.*